

Generation Interconnection System Impact Study Report

For Queue Project Q20-1405

Hard Rock 138 kV

July 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1405. The Interconnection Customer proposes a 72.6 MW solar plus storage facility interconnecting at the Hard Rock 138 kV substation in the Virginia area. The request was received on February 13, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 453 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,602,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EAST LONGMEADOW / HARD ROCK 138kV	PARSONS TAP / E LONGMEADOW SUBSTATION 138kV	104.9	225	39.7
SANGER / MYRTLE 138kV	YARNEN / MCCARTNEY 345kV	104.1	230	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EAST LONGMEADOW / HARD ROCK 138kV Line	Reconductor line	\$2,210,000	34% (\$740,000)
MYRTLE / SANER 138kV	Reconductor line	\$7,944,000	11% (\$862,000)
	Total allocated to project		\$1,602,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.